

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation. CO (BL4, BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 50

Annexure A

MCQ

Code of Conduct:

I Atchayachandran R ¹⁹²⁵²¹³⁹⁴ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Atchay

Annexure A

MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash? A) Avalanche effect ~~B) Determinism~~ C) Collision resistance D) Diffusion
2. What is the primary purpose of SHA-3? A) Faster encryption ~~B) Alternative secure hashing standard~~ C) Data compression D) Key exchange
3. Which attack tries all possible inputs to match a hash? A) Replay attack ~~B) Brute-force attack~~ C) MITM attack D) DoS attack
4. What is a salted hash? A) Encrypted hash ~~B) Hash with random data added~~ C) Shortened hash D) Reversed hash
5. Which algorithm is widely used for password hashing? A) AES ~~B) RSAC~~ ~~C) bcrypt~~ D) DES
6. What does second pre-image resistance ensure? A) Same input produces same output ~~B) Hard to find different input with same hash~~ C) Fast hashing D) Small hash size
7. What is the output size of SHA-512? A) 256 bits ~~B) 128 bits~~ ~~C) 512 bits~~ D) 1024 bits
8. Which technique slows down brute-force attacks? A) Encryption ~~B) Key exchange~~ ~~C) Key stretching~~ D) Compression
9. What is the purpose of a nonce? A) Encryption ~~B) Prevent replay attacks~~ C) Compress data D) Generate keys
10. Which function is irreversible? A) Encryption ~~B) Hashing~~ C) Compression D) Encoding
11. Which algorithm is used in DSS? A) AES ~~B) DSAC~~ ~~C) DES~~ D) RC4
12. What is required to verify a digital signature? A) Private key ~~B) Public key~~ C) Session key D) Shared key
13. What does a digital signature guarantee? A) Confidentiality ~~B) Integrity and authenticity~~ C) Compression D) Speed
14. What happens if message digest changes? A) Signature remains valid ~~B) Verification fails~~ C) Encryption fails D) Speed increases
15. Which attack involves forging a signature? A) Replay ~~B) Forgery~~ C) DoS D) Sniffing
16. What is the role of a hash in signatures? A) Encrypt data ~~B) Reduce size~~ ~~C) Ensure integrity~~ D) Store keys
17. Which key is kept secret in signatures? A) Public key ~~B) Private key~~ C) Session key D) Shared key
18. What ensures trust in certificates? A) Encryption ~~B) Certificate Authority~~ C) Compression D) Hash size
19. What is certificate revocation? A) Key generation ~~B) Canceling validity~~ C) Encryption D) Compression
20. What is OCSP used for? A) Encryption ~~B) Certificate validation~~ C) Compression D) Routing
21. What is the main goal of Kerberos tickets? A) Encryption ~~B) Authentication~~ C) Compression D) Routing
22. Which protocol uses tickets for authentication? A) SSL ~~B) Kerberos~~ C) FTP D) HTTP
23. What is a session key? A) Permanent key ~~B) Temporary key~~ C) Public key D) Private key
24. What is mutual authentication? A) One-way authentication ~~B) Both parties verify each other~~ C) Anonymous login D) Weak authentication
25. What ensures freshness in Kerberos? A) Encryption ~~B) Timestamp~~ C) Compression D) Hashing
26. What is a replay attack? A) Data modification ~~B) Resending valid data~~ C) Encryption D) Compression
27. What is the function of an authenticator in Kerberos? A) Data storage ~~B) Identity proof~~ C) Encryption D) Compression
28. What happens if system clocks differ in Kerberos? A) Faster authentication ~~B) Failure~~ C) Better security D) No effect

29. What is the KDC composed of? A) Client & Server B) AS & TGSC C) Firewall & IDS D) Router & Switch
30. What is the AS in Kerberos? A) Application Server B) Authentication Server C) Access Server D) Admin Server
31. What is X.509 certificate format used in? A) Networking B) Digital certificates C) Compression D) Routing
32. What identifies the certificate owner? A) Public key B) Subject field C) Signature D) Hash
33. What ensures certificate integrity? A) Encryption B) Digital signature C) Compression D) Routing
34. What is a root certificate? A) Expired certificate B) Self-signed trusted certificate C) Invalid certificate D) Temporary certificate
35. What is certificate pinning? A) Storing keys B) Binding certificate to host C) Encryption D) Compression
36. What is the risk of weak CA validation? A) Faster speed B) MITM attacks C) Compression D) Routing
37. What is the purpose of trust chain? A) Encryption B) Validate certificate hierarchy C) Compress data D) Routing
38. What is digital certificate used for? A) Compression B) Authentication C) Routing D) Storage
39. What ensures non-repudiation? A) Encryption B) Digital signature C) Compression D) Routing
40. What is a hash collision? A) Same input B) Same output for different inputs C) Encryption D) Compression
41. Which hash is strongest? A) MD5 B) SHA-1 C) SHA-256 D) CRC
42. What is a one-way function? A) Reversible B) Irreversible C) Fast D) Small
43. What ensures authentication in PKI? A) Encryption B) Certificates C) Compression D) Routing
44. What is key escrow? A) Key deletion B) Key storage by third party C) Encryption D) Compression
45. What is identity spoofing? A) Encryption B) Fake identity C) Compression D) Routing
46. What is trust anchor? A) Root CAB B) Client C) Server D) Router
47. What ensures integrity of message? A) Encryption B) Hash C) Compression D) Routing
48. What is digital envelope? A) Hash B) Encrypted message + key C) Compression D) Routing
49. What is certificate fingerprint? A) Encryption B) Hash of certificate C) Compression D) Routing
50. What is PKI used for? A) Routing B) Secure communication C) Compression D) Storage